A General Theory of Coolants

Thermophysical Properties, Thermohydraulic Performance, and Equivalence Classes

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Abstract

Heat is an unavoidable consequence of energy conversion, mechanical work, electrical computation, chemical reactions, biological activity, and industrial production. The ability to transport, absorb, redistribute, and reject heat is consequently a fundamental determinant of the productive capacity of the modern economy. Cooling systems are therefore not merely auxiliary engineering installations: they are part of the physical infrastructure upon which industrial production, transportation, electricity generation, computation, public health, and economic welfare depend.

This treatise develops a general mathematical and empirical framework for the study of coolants. It combines thermodynamics, heat transfer, fluid mechanics, mechanical engineering, chemical engineering, materials science, statistics, econometrics, optimisation, financial analysis, computer science, and systems engineering. Its central objectives are to derive a general representation of coolant performance, construct thermohydraulic equivalence classes, formulate economically efficient coolant-selection rules, and establish an empirical research programme based on manufacturer-published technical specifications.

The framework distinguishes sensible-heat transport from phase-change transport, coolant-side convection from total heat-exchanger performance, and intrinsic fluid properties from system-dependent engineering outcomes. It develops dimensionless heat-transfer correlations, hydraulic resistance equations, dynamic models, multi-objective optimisation, uncertainty analysis, and a statistical methodology for comparing commercial formulations.

Particular attention is given to industrial heat-transfer fluids, automotive and heavy-duty engine coolants, data-centre cooling fluids, dielectric immersion fluids, and phase-change materials. Manufacturer documentation from Dow, Eastman, and Fleetguard, independent product catalogues, and the Open Compute Project's emerging PG25 specifications provide the initial empirical foundation.

The principal theoretical proposition is that chemically different coolants may belong to the same thermohydraulic equivalence class under a specified geometry, operating regime, and set of boundary conditions, even though they differ in corrosion protection, electrical compatibility, chemical stability, environmental burden, and lifecycle cost.

The treatise concludes by proposing a Coolant Observatory: a source-audited database and computational platform for property estimation, phase-envelope analysis, thermohydraulic optimisation, equivalence testing, and economically informed coolant selection.

The treatise ends with "The End"

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1 Introduction: Heat as an Economic Constraint

1.1 The physical foundations of production

Every productive process transforms energy, matter, or information. Such transformations are accompanied by energy transfers and, in many cases, irreversible heat generation.

Examples include:

1. Electrical resistance losses in transmission networks.
2. Heat generated by internal combustion engines.
3. Thermal loads in industrial furnaces and chemical reactors.
4. Heat generated by semiconductor switching and computation.
5. Frictional heating in mechanical equipment.
6. Metabolic heat production in biological organisms.
7. Thermal loads in batteries, fuel cells, and electric motors.

The Second Law of Thermodynamics establishes that real processes generate entropy. Consequently, the complete elimination of thermal losses is generally impossible. However, their management, redistribution, recovery, and rejection can be improved.

A useful starting point is the energy balance:

$$\frac{dE_{\mathrm{sys}}}{dt} = \dot{Q}_{\mathrm{in}} - \dot{Q}_{\mathrm{out}} + \dot{W}_{\mathrm{in}} - \dot{W}_{\mathrm{out}} + \sum_{\mathrm{in}} \dot{m} h - \sum_{\mathrm{out}} \dot{m} h. \tag{1}$$

Here, E_{sys} is the energy stored within a control volume, $\dot{Q}$ denotes heat-transfer rate, $\dot{W}$ denotes work rate, and h is specific enthalpy.

The equation is a general statement of energy conservation for an open system. It provides the foundation for the coolant theory developed in this treatise.

1.2 The economic significance of cooling

Let an industrial installation produce output Y, consume productive energy E, and incur cooling expenditure C_{cool}.

A simplified production representation is:

$$Y = F(K, L, E, T_{\text{op}}, R), \tag{2}$$

where:

- K denotes physical capital;
- L denotes labour;
- E denotes energy input;
- T_{op} denotes operating temperature;
- R denotes operational reliability.

Temperature can influence productivity through equipment derating, thermal degradation, safety constraints, reduced semiconductor performance, and forced shutdowns.

The economic problem is therefore not simply to minimise temperature. It is to select a cooling architecture that maximises productive output after accounting for cooling energy, capital expenditure, maintenance, reliability, and environmental externalities.

We may write:

$$\max_{\mathbf{x},\mathbf{u}} \left[Y(\mathbf{x}, \mathbf{u}) - C_{\text{cool}}(\mathbf{x}, \mathbf{u}) - C_{\text{failure}}(\mathbf{x}, \mathbf{u}) \right]. \tag{3}$$

The variables $\mathbf{x}$ represent design choices and $\mathbf{u}$ represents operational controls.

This is the central economic motivation for a general theory of coolants.

2 Thermodynamic Foundations

2.1 The fundamental calorimetric equation

For a steady-flow coolant operating in a single phase, the heat absorbed by the fluid is determined by its enthalpy increase.

The general calorimetric equation is:

$$\boxed{\dot{Q}_{\text{coolant}} = \dot{m} \left(h_{\text{out}} - h_{\text{in}} \right)} \tag{4}$$

where:

- $\dot{Q}_{\text{coolant}}$ is heat absorption rate in watts;
- $\dot{m}$ is mass flow rate in kilograms per second;
- h_{out} is outlet specific enthalpy;
- h_{in} is inlet specific enthalpy.

For an approximately incompressible, single-phase fluid with nearly constant specific heat capacity:

$$h_{\text{out}} - h_{\text{in}} \approx c_p (T_{\text{out}} - T_{\text{in}}). \tag{5}$$

Therefore:

$$\boxed{\dot{Q} = \dot{m} c_p \Delta T} \tag{6}$$

where:

$$\Delta T = T_{\text{out}} - T_{\text{in}}. \tag{7}$$

Equation (4) is the more general formulation. Equation (6) is a useful approximation for single-phase sensible heating or cooling.

2.2 Temperature-dependent specific heat capacity

The specific heat capacity of a coolant is generally a function of temperature, pressure, and composition:

$$c_p = c_p(T, P, \mathbf{x}), \tag{8}$$

where $\mathbf{x}$ denotes composition.
For a single-phase fluid:

$$h_{\text{out}} - h_{\text{in}} = \int_{T_{\text{in}}}^{T_{\text{out}}} c_p(T, P, \mathbf{x})\, dT \tag{9}$$

under an appropriate constant-pressure approximation.
The heat-transfer rate becomes:

$$\dot{Q} = \dot{m} \int_{T_{\text{in}}}^{T_{\text{out}}} c_p(T, P, \mathbf{x})\, dT. \tag{10}$$

This expression is preferable to imposing a constant specific heat capacity over a wide temperature interval.

2.3 Polynomial approximation and mathematical degree

For empirical estimation over a restricted temperature range, consider:

$$\dot{m}(t) = a_0 + a_1 t + a_2 t^2, \tag{11}$$
$$T_{\text{out}}(t) = b_0 + b_1 t + b_2 t^2, \tag{12}$$
$$T_{\text{in}}(t) = c_0 + c_1 t + c_2 t^2. \tag{13}$$

Then:

$$\Delta T(t) = d_0 + d_1 t + d_2 t^2, \tag{14}$$

where:

$$d_i = b_i - c_i. \tag{15}$$

If c_p is constant, the instantaneous heat-transfer rate is:

$$\dot{Q}(t) = c_p(a_0 + a_1 t + a_2 t^2)(d_0 + d_1 t + d_2 t^2). \tag{16}$$

The product of two quadratic polynomials is a polynomial of degree at most four.
Consequently:

$$\deg(\dot{Q}) \leq 4. \tag{17}$$

The cumulative thermal energy absorbed is:

$$Q(t) = \int_0^t \dot{Q}(s)\, ds. \tag{18}$$

It follows that:

$$\deg(Q) \leq 5. \tag{19}$$

Thus the cumulative energy is at most quintic under these assumptions.

If specific heat capacity is approximated by a linear function of time:

$$c_p(t) = e_0 + e_1 t, \tag{20}$$

the instantaneous heat-transfer rate becomes at most quintic and the cumulative thermal energy becomes at most sextic.

This is a mathematical approximation, not a claim that specific heat capacity is intrinsically a function of time. Its time dependence arises through the temperature and composition trajectories.

2.4 Phase-change cooling

For phase-change cooling, the sensible-heat equation alone is insufficient.

For a simple vaporisation process:

$$Q = mL_v, \tag{21}$$

where L_v is latent heat of vaporisation.

If the fluid is first heated to saturation and subsequently vaporised:

$$Q = m \left[\int_{T_{\text{in}}}^{T_{\text{sat}}} c_p(T)\, dT + L_v \right]. \tag{22}$$

The general enthalpy formulation automatically accommodates phase changes, provided the appropriate equilibrium or nonequilibrium thermodynamic model is used.

3 The Thermophysical Property Vector

3.1 The principal properties

The first-order thermophysical representation of a coolant is:

$$\boxed{\mathbf{X} = \begin{pmatrix} c_p \\ k \\ \rho \\ \mu \end{pmatrix}} \tag{23}$$

where:

- c_p: specific heat capacity;
- k: thermal conductivity;
- ρ: density;
- μ: dynamic viscosity.

These properties are functions of the thermodynamic state:

$$\mathbf{X} = \mathbf{X}(T, P, \mathbf{x}, \mathcal{A}), \tag{24}$$

where $\mathcal{A}$ represents the additive and inhibitor package.

The property vector is not a complete specification of coolant suitability. Chemical compatibility, electrical properties, toxicity, flammability, and stability must be considered separately.

3.2 Specific heat capacity

Specific heat capacity measures the energy required to raise the temperature of a unit mass by one kelvin.

$$c_p = \left(\frac{\partial h}{\partial T} \right)_{P,\mathbf{x}}. \tag{25}$$

For a specified mass and temperature rise:

$$Q = mc_p \Delta T \tag{26}$$

under constant-c_p approximation.

A higher c_p therefore increases sensible heat storage per unit mass, all else equal.

3.3 Thermal conductivity

Thermal conductivity governs conductive heat transport through a material.

Fourier's law is:

$$\mathbf{q} = -k \nabla T. \tag{27}$$

For one-dimensional steady conduction through a plane wall:

$$\dot{Q} = kA \frac{T_1 - T_2}{L}. \tag{28}$$

A higher thermal conductivity reduces conductive thermal resistance for a fixed geometry and temperature difference.

3.4 Density

Density is:

$$\rho = \frac{m}{V}. \tag{29}$$

It affects mass flow, volumetric flow, inertia, Reynolds number, hydrostatic pressure, and hydraulic energy requirements.

Its effect depends on the operating constraint.

At fixed volumetric flow:

$$\dot{m} = \rho \dot{V}. \tag{30}$$

At fixed mass flow:

$$\dot{V} = \frac{\dot{m}}{\rho}. \tag{31}$$

Consequently, density cannot be assigned a universal positive or negative performance weight without specifying the system boundary and operating conditions.

3.5 Viscosity

Dynamic viscosity measures resistance to shear deformation.

For a Newtonian fluid:

$$\tau = \mu \frac{du}{dy}. \tag{32}$$

Kinematic viscosity is:

$$\nu = \frac{\mu}{\rho}. \tag{33}$$

When a manufacturer reports kinematic viscosity, dynamic viscosity can be recovered through:

$$\boxed{\mu = \rho \nu} \tag{34}$$

provided the measurements refer to the same temperature and composition and the units are consistent.

3.6 Additional properties

For a comprehensive coolant model, the property vector should be expanded:

$$\mathbf{X}^* = \begin{pmatrix} c_p & k & \rho & \mu & P_v & \sigma & \epsilon_r & L_v & \beta \end{pmatrix}^T, \tag{35}$$

where:

- P_v is vapour pressure;
- σ is electrical conductivity;
- ϵ_r is relative permittivity;
- L_v is latent heat of vaporisation;
- β is volumetric thermal expansion coefficient.

Other important descriptors include freezing point, boiling point, flash point, pour point, dielectric strength, corrosivity, chemical stability, and environmental persistence.

4 Heat Exchanger Theory

4.1 The fundamental heat-transfer equation

For a heat exchanger:

$$\boxed{\dot{Q} = UA\Delta T_{\text{lm}}} \tag{36}$$

where U is the overall heat-transfer coefficient, A is the reference heat-transfer area, and ΔT_{lm} is the logarithmic mean temperature difference.

For terminal temperature differences ΔT_1 and ΔT_2:

$$\Delta T_{\text{lm}} = \frac{\Delta T_1 - \Delta T_2}{\ln(\Delta T_1 / \Delta T_2)}. \tag{37}$$

The equation is applicable under the relevant steady-state heat-exchanger assumptions. Multipass and cross-flow arrangements may require correction factors or more detailed exchanger models.

4.2 Coupling coolant calorimetry and heat transfer

The heat absorbed by the coolant must correspond to the heat transferred through the exchanger, after accounting for storage, losses, and measurement discrepancies.

For an ideal steady-state exchanger:

$$\boxed{\dot{m}(h_{\text{out}} - h_{\text{in}}) = UA\Delta T_{\text{lm}}} \tag{38}$$

For constant specific heat capacity:

$$\dot{m}c_p(T_{\text{out}} - T_{\text{in}}) = UA\Delta T_{\text{lm}}. \tag{39}$$

A measurement residual can be defined as:

$$\varepsilon(t) = \dot{Q}_{\text{coolant}}(t) - \dot{Q}_{\text{exchanger}}(t). \tag{40}$$

Persistent nonzero residuals may indicate transient thermal storage, heat leakage, sensor bias, flow maldistribution, or model misspecification.

4.3 Overall heat-transfer resistance

A simplified series-resistance model is:

$$\frac{1}{U} = \frac{1}{h_i} + R_{\text{wall}} + R_{\text{fouling}} + \frac{1}{h_o}. \tag{41}$$

The terms must be converted to a common area basis.
For a cylindrical wall:

$$R_{\text{wall}} = \frac{\ln(r_o/r_i)}{2\pi k_w L}. \tag{42}$$

This establishes an important engineering principle: improving the coolant-side heat-transfer coefficient may yield limited system improvement if wall conduction, fouling, or the opposing fluid-side convection is the dominant resistance.

5 Fluid Mechanics and Hydraulic Resistance

5.1 Continuity

For a steady incompressible flow:

$$\dot{m} = \rho A_c v, \tag{43}$$

where A_c is cross-sectional flow area and v is mean velocity.
For a circular pipe:

$$A_c = \frac{\pi D^2}{4}. \tag{44}$$

Therefore:

$$v = \frac{4\dot{m}}{\pi \rho D^2}. \tag{45}$$

5.2 Darcy–Weisbach equation

The pressure loss due to pipe friction is:

$$\boxed{\Delta P = f_D \frac{L}{D} \frac{\rho v^2}{2}} \tag{46}$$

where f_D is the Darcy friction factor.
Substituting the circular-pipe velocity expression gives:

$$\Delta P = \frac{8 f_D L \dot{m}^2}{\pi^2 \rho D^5}. \tag{47}$$

This expression assumes a circular pipe and uses the Darcy friction factor. The friction factor depends on Reynolds number and relative roughness.

5.3 Reynolds number

$$\boxed{Re = \frac{\rho v D}{\mu}} \tag{48}$$

The Reynolds number characterises the relative importance of inertial and viscous forces.

Flow regime classification is geometry-dependent and should not be inferred from a single universal threshold in all circumstances.

5.4 Pumping power

The hydraulic power delivered to the fluid is:

$$P_{\text{hyd}} = \Delta P \dot{V}. \tag{49}$$

The electrical or shaft power required by the pump is approximately:

$$\boxed{P_{\text{pump}} = \frac{\Delta P \dot{V}}{\eta_p}} \tag{50}$$

where η_p is the relevant pump efficiency.

For a circular pipe under simplified fixed-property conditions:

$$P_{\text{pump}} = \frac{8 f_D L \dot{m}^3}{\pi^2 \rho^2 D^5 \eta_p}. \tag{51}$$

This expression suggests cubic mass-flow scaling when friction factor, density, geometry, and pump efficiency are held constant.

In actual systems, friction factor and pump efficiency vary with operating conditions. The cubic relationship is therefore a conditional approximation, not a universal law.

5.5 System-level pressure losses

A practical model must include:

$$\Delta P_{\text{system}} = \Delta P_{\text{pipe}} + \Delta P_{\text{fittings}} + \Delta P_{\text{valves}} + \Delta P_{\text{HX}} + \Delta P_{\text{coldplate}}. \tag{52}$$

The pump operating point is determined by the intersection of the pump curve and system resistance curve.

This is especially important in data-centre cooling systems, where cold plates, quick disconnects, manifolds, and control valves may contribute materially to total hydraulic resistance.

6 Dimensionless Heat-Transfer Theory

6.1 Prandtl number

$$\boxed{Pr = \frac{\mu c_p}{k}} \tag{53}$$

The Prandtl number relates momentum diffusivity to thermal diffusivity.

6.2 Nusselt number

$$\boxed{Nu = \frac{hD}{k}} \tag{54}$$

The Nusselt number represents dimensionless convective heat transfer.

6.3 General empirical correlation

A common correlation form is:

$$\boxed{Nu = C Re^m Pr^n} \tag{55}$$

The constants depend on geometry, flow regime, boundary conditions, and thermal development.

For turbulent flow in a smooth circular tube, the Dittus–Boelter correlation is:

$$Nu = 0.023 Re^{0.8} Pr^n, \tag{56}$$

with commonly used exponents near 0.4 for fluid heating and 0.3 for fluid cooling within the correlation's applicable range.

Other correlations, including Gnielinski and Sieder–Tate, may be preferable under different operating conditions.

6.4 Derivation of a property-based heat-transfer expression

From:

$$Nu = \frac{hD}{k} = CRe^m Pr^n, \tag{57}$$

we obtain:

$$h = \frac{k}{D} CRe^m Pr^n. \tag{58}$$

Substituting the dimensionless definitions:

$$h = \frac{k}{D} C \left(\frac{\rho v D}{\mu}\right)^m \left(\frac{\mu c_p}{k}\right)^n \tag{59}$$

$$= C D^{m-1} \rho^m v^m \mu^{n-m} c_p^n k^{1-n}. \tag{60}$$

For fixed geometry and velocity:

$$\boxed{h \propto \rho^m \mu^{n-m} c_p^n k^{1-n}} \tag{61}$$

This expression provides a useful theoretical basis for comparing coolant properties.
It is not a universal performance law. It is conditional on the selected correlation and its validity.

7 Thermohydraulic Performance

7.1 A preliminary efficiency measure

A useful preliminary measure is:

$$\mathcal{E}_{\mathrm{TH}} = \frac{\dot{Q}}{P_{\mathrm{pump}}}. \tag{62}$$

It measures transported thermal power per unit pumping power.
It should not be confused with the coefficient of performance of a refrigeration or heat-pump system.
A dimensionless relative index is:

$$\boxed{\Phi_{\mathrm{TH}} = \frac{(\dot{Q}/P_{\mathrm{pump}})_{\mathrm{candidate}}}{(\dot{Q}/P_{\mathrm{pump}})_{\mathrm{reference}}}} \tag{63}$$

The candidate and reference must be compared under clearly defined, equivalent service conditions.

7.2 Why a single property index is insufficient

A raw property score such as:

$$\Phi = \left(\frac{\rho}{\rho_0}\right)^2 \left(\frac{c_p}{c_{p0}}\right) \left(\frac{k}{k_0}\right) \left(\frac{\mu_0}{\mu}\right) \tag{64}$$

has no universal physical justification.

The density exponent depends on the selected flow constraint and hydraulic model. The other exponents also depend on geometry and operating regime.

A scientifically defensible index should be derived from system outcomes rather than arbitrary property weights.

7.3 A multi-objective formulation

Let the decision vector be:

$$\mathbf{z} = \begin{pmatrix} \dot{m} & D & A & \mathbf{x} & \mathbf{u} \end{pmatrix}^T. \tag{65}$$

A general optimisation problem is:

$$\max_{\mathbf{z}} \begin{pmatrix} \dot{Q}(\mathbf{z}) \\ -P_{\text{system}}(\mathbf{z}) \\ -C_{\text{lifecycle}}(\mathbf{z}) \\ R_{\text{reliability}}(\mathbf{z}) \end{pmatrix} \tag{66}$$

subject to:

$$T_{\text{system}}(t) \leq T_{\max}, \tag{67}$$
$$\Delta P_{\text{system}}(t) \leq \Delta P_{\max}, \tag{68}$$
$$\dot{Q}(t) \geq \dot{Q}_{\text{required}}(t), \tag{69}$$
$$\mathbf{X}(t) \in \mathcal{X}_{\text{feasible}}, \tag{70}$$
$$(T(t), P(t), \mathbf{x}(t)) \in \Omega_{\text{phase}}, \tag{71}$$
$$\mathbf{x}(t) \in \Omega_{\text{compatibility}}. \tag{72}$$

The feasible set must also incorporate applicable safety, material, regulatory, and equipment-vendor constraints.

The solution is generally a Pareto frontier rather than one universally optimal coolant.

8 Thermohydraulic Equivalence Classes

8.1 Definition

Consider two coolant formulations:

$$\mathbf{X}_1 = (c_{p1}, k_1, \rho_1, \mu_1)^T \tag{73}$$

and:

$$\mathbf{X}_2 = (c_{p2}, k_2, \rho_2, \mu_2)^T. \tag{74}$$

They are thermohydraulically equivalent under operating condition $\mathcal{C}$ if they produce equivalent specified thermal and hydraulic outcomes within a predefined tolerance.

The condition $\mathcal{C}$ includes:

- Geometry.
- Flow constraint.
- Temperature and pressure.
- Heat-transfer boundary conditions.
- Surface roughness.
- Pump characteristics.
- Exchanger configuration.

Equivalence is therefore conditional, not absolute.

8.2 Thermal equivalence

For equal exchanger area and equal temperature driving force:

$$U_1 = U_2 \tag{75}$$

is a sufficient condition for equal heat-transfer rates in the simplified exchanger model.

More generally:

$$U_1 A_1 \Delta T_{\mathrm{lm},1} = U_2 A_2 \Delta T_{\mathrm{lm},2}. \tag{76}$$

8.3 Hydraulic equivalence

Under a specified geometry and flow condition:

$$\Delta P_1 = \Delta P_2 \tag{77}$$

defines pressure-drop equivalence.

If pump efficiencies are different, equal pressure drop does not necessarily imply equal electrical pumping power.

The more complete requirement is:

$$P_{\mathrm{pump},1} = P_{\mathrm{pump},2}. \tag{78}$$

8.4 A preliminary equivalence manifold

Under a common heat-transfer correlation and fixed geometry and velocity:

$$h \propto \rho^m \mu^{n-m} c_p^n k^{1-n}. \tag{79}$$

Define:

$$\mathbf{z} = \begin{pmatrix} \ln \rho \\ \ln \mu \\ \ln c_p \\ \ln k \end{pmatrix} \tag{80}$$

and:

$$\mathbf{w} = \begin{pmatrix} m \\ n-m \\ n \\ 1-n \end{pmatrix}. \tag{81}$$

Taking logarithms:

$$\ln h = \ln C + (m-1)\ln D + m \ln \rho + (n-m)\ln \mu + n \ln c_p + (1-n)\ln k + m \ln v. \tag{82}$$

For identical geometry and velocity, equal convective coefficients imply:

$$\boxed{\mathbf{w}^T(\mathbf{z}_1 - \mathbf{z}_2) = 0.} \tag{83}$$

8.5 Proposition 1: local dimensionality of the thermal equivalence set

Suppose the correlation exponents are fixed and the vector $\mathbf{w}$ is nonzero. Then the set of log-property vectors producing the same convective heat-transfer coefficient is a hyperplane of codimension one in four-dimensional log-property space.

Proof.

The equality condition is one nontrivial linear restriction:

$$\mathbf{w}^T \mathbf{z} = c. \tag{84}$$

A single independent linear restriction on a four-dimensional vector space leaves a three-dimensional affine subspace.

Therefore the thermal equivalence set is locally three-dimensional.

□

8.6 Proposition 2: hydraulic constraints reduce the equivalence set

If an independent hydraulic equivalence condition is imposed, the admissible set is generally reduced further.

Proof.

Let the hydraulic condition be represented locally by:

$$g(\mathbf{z}) = 0. \tag{85}$$

If its gradient is linearly independent of the thermal constraint gradient, it introduces another independent restriction.

The local dimension is then reduced by one additional degree of freedom.

If the constraints are dependent, the dimension is not necessarily reduced.

□

This result explains why a single heat-transfer correlation cannot establish complete coolant equivalence.

8.7 Equivalence as a relation

Define a tolerance-based equivalence relation on a set of candidate fluids under fixed conditions:

$$i \sim_\epsilon j \tag{86}$$

if the relevant thermal and hydraulic performance differences fall within specified tolerances.

For mathematical consistency, a symmetric tolerance rule is preferable, for example:

$$d(i, j) \leq \epsilon, \tag{87}$$

where d is a properly defined distance between normalised outcome vectors.

A simple threshold rule need not be transitive. Consequently, equivalence classes should be constructed using a clustering or graph-connectivity method that explicitly handles this issue.

9 Phase Equilibrium and Chemical Stability

9.1 Pure-fluid phase behaviour

For a pure substance, phase equilibrium between liquid and vapour is characterised by equality of chemical potentials:

$$\mu_{\mathrm{chem},l}(T, P) = \mu_{\mathrm{chem},v}(T, P). \tag{88}$$

The subscript distinguishes chemical potential from dynamic viscosity.

A phase diagram identifies regions of thermodynamic stability.

Important boundaries include:

- Liquid–vapour coexistence.
- Solid–liquid coexistence.
- Solid–vapour coexistence.
- Critical point.
- Triple point.

9.2 Mixture phase envelopes

For a coolant mixture:

$$\mathcal{P} = \mathcal{P}(T, P, \mathbf{x}). \tag{89}$$

Composition affects freezing, boiling, vapour-liquid equilibrium, and other phase transitions.

For mixtures, bubble-point and dew-point curves may be more appropriate than a single boiling temperature.

9.3 Operating envelope

Define the permitted operating region:

$$\Omega = \{(T, P, \mathbf{x}) : \text{all specified thermodynamic and safety constraints hold}\}. \tag{90}$$

A valid operating trajectory must satisfy:

$$(T(t), P(t), \mathbf{x}(t)) \in \Omega \qquad \forall t. \tag{91}$$

A practical design should maintain appropriate margins from freezing, boiling, cavitation, decomposition, and other failure boundaries.

9.4 Chemical stability is not phase equilibrium

Thermal degradation and chemical decomposition are not ordinary equilibrium phase boundaries.

They should be represented through additional state constraints:

$$D_{\text{chem}}(t) \leq D_{\max}, \tag{92}$$

where D_{chem} is a degradation indicator.

A simple kinetic approximation is:

$$\frac{dD_{\text{chem}}}{dt} = k_0 \exp\left(-\frac{E_a}{RT}\right) f(\mathbf{x}, O_2, \text{contaminants}). \tag{93}$$

This expression is a generic Arrhenius-type representation. Actual coolant degradation kinetics must be identified experimentally for each formulation.

10 Chemistry, Materials Science, and Corrosion

10.1 Chemical formulation

Commercial coolants are rarely pure substances. They commonly contain:

- Water or glycol base fluids.
- Corrosion inhibitors.
- Buffering agents.
- Antifoaming agents.
- Dyes.
- Biocides or other specialised additives.

Two formulations with the same base-fluid concentration may therefore have different chemical behaviour.

10.2 Corrosion and materials compatibility

A coolant must be compatible with all wetted materials.

Potential materials include:

- Copper.
- Aluminium.
- Stainless steel.
- Carbon steel.
- Nickel alloys.
- Elastomers.
- Plastics.
- Solder and brazing materials.

Corrosion is influenced by temperature, pH, dissolved oxygen, ionic composition, flow velocity, galvanic coupling, and inhibitor concentration.

A useful empirical corrosion response is:

$$r_{\mathrm{corr}} = f(T, P, \mathrm{pH}, \sigma, \mathbf{x}, \mathcal{M}), \tag{94}$$

where $\mathcal{M}$ represents the wetted material configuration.

The function must be established through relevant corrosion tests rather than inferred from heat-transfer properties.

10.3 Electrical compatibility

For data-centre and battery applications, electrical conductivity and dielectric strength may be critical.

A fluid suitable for a conventional liquid cooling loop may not be suitable for direct contact with electrical components.

The coolant selection problem must therefore distinguish:

1. Single-phase indirect cooling.
2. Single-phase direct-to-chip cooling.
3. Immersion cooling.
4. Two-phase dielectric cooling.

The relevant electrical and material requirements differ substantially.

11 Biological and Human Factors

11.1 Biological heat management

Living organisms regulate temperature through mechanisms including convection, conduction, radiation, evaporation, and physiological control.

A simplified biological heat balance is:

$$\dot{S} = M - W - E - R - C - K, \tag{95}$$

where:

- $\dot{S}$ is body heat storage rate;
- M is metabolic heat production;
- W is external mechanical work;

- E is evaporative heat loss;
- R is radiative heat loss;
- C is convective heat loss;
- K is conductive heat loss.

This is not a coolant equation, but it illustrates the wider importance of heat-transfer engineering to biological systems.

11.2 Occupational health

Industrial coolant selection must account for worker exposure through skin contact, inhalation of aerosols, accidental ingestion, leaks, and maintenance operations.

Safety data sheets, exposure limits, handling procedures, ventilation, personal protective equipment, and disposal requirements must be considered.

The thermohydraulic optimum is not automatically the occupational-health optimum.

11.3 Psychology and human factors

Cooling-system operation depends on human decisions, maintenance behaviour, alarm interpretation, and organisational procedures.

Poor interface design or alarm overload can delay response to overheating, leakage, or pump failure.

A human-factors model may represent failure probability as:

$$P(F) = f(W_{\text{load}}, A_{\text{alarm}}, T_{\text{training}}, C_{\text{procedure}}), \tag{96}$$

where the arguments represent workload, alarm architecture, training, and procedural quality.

Psychiatric diagnosis is not a variable in coolant thermodynamics. Mental-health considerations become relevant only where occupational conditions, fatigue, stress, or safety-critical work practices are being assessed through appropriate evidence.

12 Economic Theory of Cooling

12.1 Cooling as an intermediate productive input

Cooling services are intermediate inputs into industrial and digital production.

Let output be:

$$Y = F(K, L, E, H), \tag{97}$$

where H denotes useful cooling service.

The marginal product of cooling is:

$$MP_H = \frac{\partial Y}{\partial H}. \tag{98}$$

Where overheating constrains production, the marginal product of reliable cooling can be substantial.

12.2 Cost function

Define total lifecycle cost:

$$C_{\text{life}} = C_{\text{fluid}} + C_{\text{capital}} + C_{\text{pump}} + C_{\text{maintenance}} + C_{\text{replacement}} + C_{\text{failure}} + C_{\text{disposal}}. \tag{99}$$

The relevant decision is to minimise lifecycle cost subject to thermal service and safety requirements.

$$\min_{\mathbf{z}} C_{\text{life}}(\mathbf{z}) \tag{100}$$

subject to:

$$\dot{Q}(\mathbf{z}) \geq \dot{Q}_{\text{required}}, \tag{101}$$
$$T_{\max}(\mathbf{z}) \leq T_{\text{allowable}}, \tag{102}$$
$$P_{\text{system}}(\mathbf{z}) \leq P_{\text{budget}}, \tag{103}$$
$$\mathbf{z} \in \mathcal{F}. \tag{104}$$

12.3 Net present value

Let annual cash flows associated with a cooling investment be CF_t, initial investment be I_0, and discount rate be r.

Then:

$$NPV = -I_0 + \sum_{t=1}^{N} \frac{CF_t}{(1+r)^t}. \tag{105}$$

The cash flows may include:

- Electricity savings.
- Increased productive uptime.
- Reduced maintenance.
- Lower fluid replacement expenditure.
- Avoided equipment failures.
- Environmental compliance costs.

The discount rate should reflect the financing structure, risk, and relevant opportunity cost of capital.

12.4 Levelised cost of cooling

A useful comparison measure is:

$$LCOC = \frac{PV(\text{all cooling costs})}{PV(\text{useful thermal energy removed})}. \tag{106}$$

This permits comparison of cooling systems with different capital costs, service lives, and operating profiles.

12.5 Welfare economics

Private cost minimisation does not necessarily maximise social welfare.

Let:

$$W = B_{\text{production}} - C_{\text{private}} - C_{\text{environment}} - C_{\text{health}} - C_{\text{systemic}}. \tag{107}$$

Environmental burdens may include greenhouse-gas emissions, water consumption, chemical release, and end-of-life disposal.

A socially efficient selection may therefore differ from the privately cheapest coolant.

12.6 Warfare economics and critical infrastructure

Thermal management is relevant to national infrastructure resilience, including electricity grids, transport systems, hospitals, communications, and computing facilities.

The economic objective is to preserve essential services under disruption.

A resilience-adjusted cost function may include:

$$C_{\text{resilience}} = C_{\text{normal}} + \mathbb{E}[C_{\text{disruption}}] + C_{\text{redundancy}}. \tag{108}$$

This is a civil infrastructure and resilience framework. It does not imply that cooling technology should be designed as an offensive weapon.

For critical infrastructure, the appropriate emphasis is continuity of service, safety, redundancy, repairability, and protection of civilian welfare.

13 Statistics and Econometrics

13.1 The empirical unit of observation

The primary observation should be a product–composition–temperature–pressure record.

A dataset row is not simply a product name. It must identify the precise formulation and thermodynamic state.

Let:

$$D = \{i, j, T, P, \mathbf{x}, c_p, k, \rho, \mu, P_v, \sigma\}\,. \tag{109}$$

The data structure must distinguish reported values from interpolated or estimated values.

13.2 Temperature-dependent models

Possible starting specifications include:

$$c_p(T) = a_0 + a_1 T + a_2 T^2, \tag{110}$$

$$k(T) = b_0 + b_1 T + b_2 T^2, \tag{111}$$

$$\rho(T) = c_0 + c_1 T + c_2 T^2, \tag{112}$$

$$\ln \mu(T) = d_0 + \frac{d_1}{T_K}. \tag{113}$$

These are candidate functional forms, not universal constitutive laws.

Viscosity often requires nonlinear temperature models. Model choice should be determined by residual analysis and predictive validation.

13.3 Statistical model comparison

For each fitted model, report:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{114}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{115}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}. \tag{116}$$

Model selection may also use AIC and BIC:

$$AIC = 2p - 2\ln \hat{L}, \tag{117}$$

$$BIC = p \ln n - 2 \ln \hat{L}. \tag{118}$$

The selected model should also be tested out of sample.

13.4 Hierarchical modelling

Products are nested within manufacturers and chemical families.

A hierarchical model can account for these structures:

$$y_{ij} = \beta_0 + \beta_1 T_{ij} + \beta_2 T_{ij}^2 + u_j + \epsilon_{ij}, \tag{119}$$

where u_j is a manufacturer- or formulation-level random effect.

More elaborate models may include concentration, pressure, additive chemistry, and interactions.

13.5 Measurement uncertainty

Suppose the measured heat-transfer rate is:

$$\dot{Q} = \dot{m} c_p \Delta T. \tag{120}$$

For approximately independent measurement errors, first-order uncertainty propagation gives:

$$\left(\frac{u_{\dot{Q}}}{\dot{Q}}\right)^2 \approx \left(\frac{u_{\dot{m}}}{\dot{m}}\right)^2 + \left(\frac{u_{c_p}}{c_p}\right)^2 + \left(\frac{u_{\Delta T}}{\Delta T}\right)^2. \tag{121}$$

If errors are correlated, covariance terms must be included.

A coolant ranking should not be reported as decisive where measurement uncertainty is larger than the performance difference.

13.6 Causal interpretation

Manufacturer specifications are observational product data, not randomised experiments.

A correlation between viscosity and reported performance does not establish a causal effect unless geometry, flow, temperature, pressure, and other confounders are appropriately controlled.

The empirical programme should distinguish:

1. Descriptive comparison.
2. Predictive modelling.
3. Engineering simulation.
4. Controlled experimental validation.
5. Causal inference.

These are different evidential levels.

14 Initial Manufacturer Evidence

14.1 Dow DOWFROST HD

Dow publishes temperature-dependent thermophysical properties for DOWFROST HD at specified propylene-glycol concentrations.

The following table reproduces selected values for the 40 percent propylene-glycol-by-volume formulation.

Table 1: DOWFROST HD at 40 percent propylene glycol by volume

Temperature °C	c_p kJ/(kg K)	ρ kg/m^3	k W/(m K)	μ mPa s
-20	3.453	1066.76	0.3635	48.9043
10	3.564	1055.38	0.3936	7.2173
40	3.675	1039.77	0.4150	2.2389
65	3.767	1023.55	0.4262	1.1762
90	3.859	1004.39	0.4313	0.7462
120	3.970	977.53	0.4294	0.5084

The data demonstrate that the fluid's viscosity changes substantially over the reported temperature range. Density declines with temperature, while specific heat capacity rises.

Thermal conductivity is not strictly monotonic across the entire table.

The concentration basis must be preserved. The 40 percent volume formulation cannot be silently combined with 40 percent by mass data.

14.2 Eastman Therminol 75

Eastman publishes temperature-dependent properties for Therminol 75.

Table 2: Selected Therminol 75 observations

Temperature °C	ρ kg/m^3	c_p kJ/(kg K)	k W/(m K)	μ mPa s	P_v kPa
71	1050	1.68	0.1314	5.07	0.014
80	1040	1.71	0.1307	4.29	0.022
90	1030	1.74	0.1300	3.60	0.038
100	1030	1.77	0.1292	3.04	0.063

The values illustrate why coolant comparisons must be application-specific.

A high-temperature synthetic heat-transfer fluid should not be ranked against an aqueous data-centre coolant without accounting for operating temperature, system geometry, heat-transfer duty, vapour pressure, and application constraints.

14.3 Fleetguard

Fleetguard's product catalogue identifies several engine coolant formulations, including ES Compleat OAT and ES Compleat products.

The ES Compleat OAT product description identifies an ethylene-glycol concentrate and an organic additive technology formulation. Fleetguard also identifies premixed 50/50 variants.

These products are relevant to the automotive and heavy-duty engine cohort.

However, the product catalogue alone does not provide a complete temperature-dependent dataset for c_p, k, ρ, and μ.

The empirical database must therefore distinguish product identification from quantitative thermophysical observations.

14.4 Data-centre cooling and PG25

The Open Compute Project's Cooling Environments programme includes a 2026 base specification for single-phase propylene-glycol-based cold-plate coolants.

The specification provides a common technical reference for the PG25 category.

The Coolant.com supplier directory identifies manufacturers and product families offering PG25 fluids, including BASF, Shell, Castrol, Dow, Dynalene, PFX Group, Ecolab, and Arteco.

These directory entries are useful for product discovery. They are not substitutes for manufacturer technical data sheets or independent measurements.

14.5 Initial evidence matrix

Table 3: Initial empirical research cohorts

Cohort	Examples	Primary research objective
Industrial fluids	DOWFROST, Therminol	Temperature-dependent property modelling
Engine coolants	Fleetguard ES Compleat	Engine operating envelope and materials compatibility
Data-centre fluids	PG25 formulations	Cross-manufacturer thermohydraulic comparison
Dielectric fluids	Immersion fluids	Thermal and electrical compatibility
Phase-change fluids	Refrigerants and phase-change media	Latent-heat and phase-equilibrium modelling

15 Database Architecture and Data Governance

15.1 Core database schema

Each observation should contain the following fields.

Table 4: Proposed coolant database

Field	Description
Product ID	Unique product identifier
Manufacturer	Legal manufacturer or supplier
Product name	Exact commercial product designation
Chemical family	Glycol, hydrocarbon, ester, dielectric, etc.
Concentration	Exact composition and concentration basis
Temperature	Measurement temperature
Pressure	Measurement pressure
Specific heat	c_p, with units
Conductivity	k, with units
Density	ρ, with units
Dynamic viscosity	μ, with units
Kinematic viscosity	ν, with units
Vapour pressure	P_v
Phase limits	Freezing, boiling, bubble and dew information
Electrical properties	Conductivity and dielectric strength
Stability	Degradation and ageing information
Compatibility	Wetted-material compatibility
Source document	Original manufacturer document
Revision date	Publication or revision date
Measurement method	ASTM, ISO, DIN or other method
Data status	Measured, typical, guaranteed, interpolated or estimated
Uncertainty	Published or estimated uncertainty

15.2 Data provenance

Every numerical observation should have an auditable source.

The database should preserve:

1. Original document.
2. Original units.
3. Original concentration basis.
4. Original temperature and pressure.

5. Document revision date.
6. Extraction method.
7. Any unit conversion.
8. Any interpolation or model estimation.

No estimated value should be presented as a manufacturer measurement.

15.3 Data quality hierarchy

A useful evidence hierarchy is:

1. Independent laboratory measurement with documented method and uncertainty.
2. Manufacturer-guaranteed specification.
3. Manufacturer-reported typical property data.
4. Manufacturer technical guidance or engineering calculator.
5. Independently compiled directory with traceable manufacturer references.
6. Secondary commentary without a traceable technical source.

The hierarchy is not a substitute for evaluating the suitability of the test method.

16 Computational Architecture

16.1 Coolant Observatory

The proposed computational system has five principal components:

1. Product and source database.
2. Thermophysical property engine.
3. Phase-equilibrium and stability engine.
4. Thermohydraulic simulation engine.
5. Optimisation and equivalence engine.

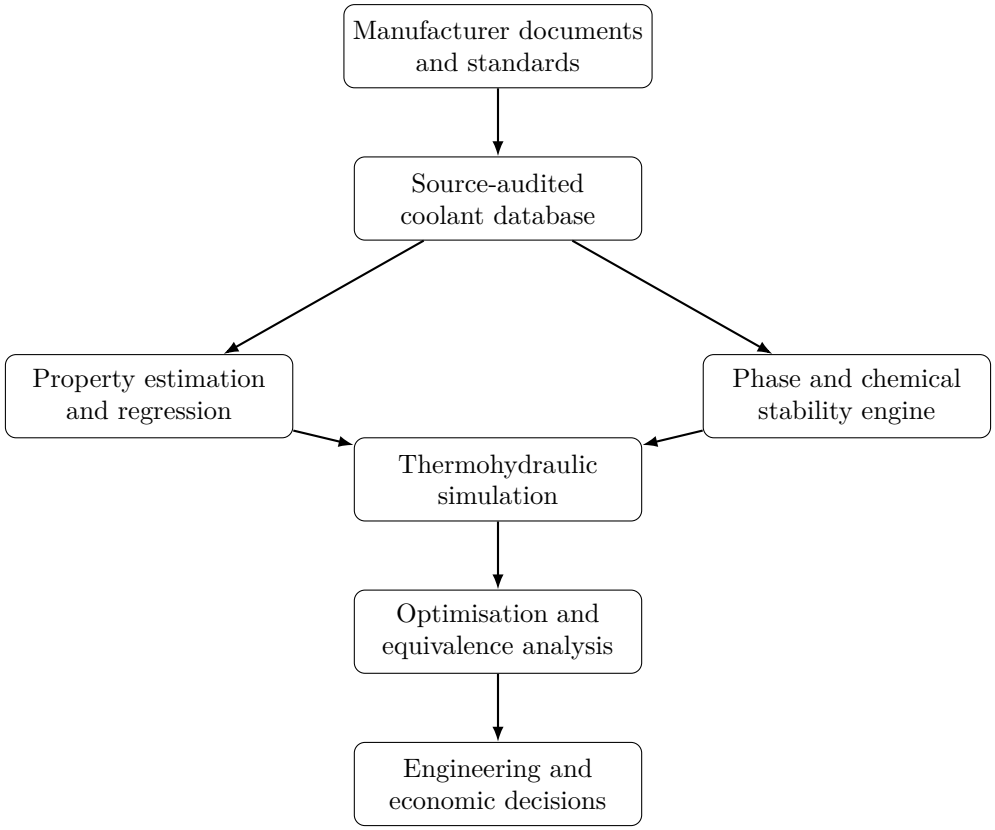


Figure 1: Proposed Coolant Observatory architecture

16.2 Algorithmic workflow

```
ALGORITHM: COOLANT-OBSERVATORY

INPUT:
    Manufacturer documents
    Product identifiers
    Thermodynamic conditions
    System geometry
    Required thermal duty
    Safety and compatibility constraints

1. EXTRACT product information.
2. RECORD original source and revision.
3. NORMALISE units without discarding original values.
4. VALIDATE concentration basis and measurement conditions.
5. CLASSIFY each observation by evidence quality.
6. FIT temperature-dependent property models.
7. VALIDATE models using residual and out-of-sample tests.
8. CONSTRUCT phase and stability envelopes.
9. CALCULATE Reynolds, Prandtl and Nusselt numbers.
10. CALCULATE heat-transfer coefficients.
11. CALCULATE pressure losses and pump power.
12. EVALUATE thermal and hydraulic constraints.
13. CALCULATE lifecycle economic measures.
14. IDENTIFY candidate equivalence groups.
15. QUANTIFY uncertainty and sensitivity.
16. EXPORT rankings, models and source audit.
```

```
OUTPUT:
    Validated candidate set
    Property models
    Phase envelopes
    Thermohydraulic performance
    Equivalence groups
    Economic comparison
    Uncertainty report
END
```

16.3 Software architecture

A practical implementation may use:

- Python for data acquisition, modelling, and optimisation.
- SQL for structured product and observation storage.
- Pandas for tabular data management.
- NumPy and SciPy for numerical computation.
- Statsmodels for econometric estimation.
- Scikit-learn for predictive modelling and clustering.
- CoolProp or NIST REFPROP for appropriate fluid-property calculations.
- Modelica or comparable simulation environments for system dynamics.
- Version control for reproducibility.

Commercial manufacturer data should remain distinguishable from property values generated by a thermodynamic library.

17 Artificial Intelligence and Machine Learning

17.1 Property prediction

A supervised model may estimate a missing property:

$$\hat{y} = f_\theta(T, P, \mathbf{x}, \mathcal{A}), \tag{122}$$

where y may represent viscosity, conductivity, density, or specific heat.

Training data should be partitioned by product or formulation, not merely by randomly selected temperature observations. Otherwise, observations from the same formulation may leak into both training and test sets.

17.2 Physics-informed learning

A physics-informed model should respect:

- Positive density.
- Positive viscosity.
- Positive specific heat capacity.
- Positive thermal conductivity.
- Thermodynamic consistency.
- Appropriate phase boundaries.

- Relevant conservation equations.

For example, positivity can be enforced by modelling:

$$\mu(T) = \exp(g_\theta(T)). \tag{123}$$

The model then cannot produce a negative viscosity.

17.3 Multi-objective learning

A machine-learning system may estimate several outputs simultaneously:

$$\hat{\mathbf{y}} = \begin{pmatrix} \hat{Q} \\ \widehat{\Delta P} \\ \hat{P}_{\text{pump}} \\ \hat{D}_{\text{chem}} \end{pmatrix}. \tag{124}$$

The model should be used as a decision-support system, not as an unverified replacement for engineering validation.

17.4 Digital twins

A digital twin of a cooling installation can combine:

1. Real-time temperature measurements.
2. Flow-rate measurements.
3. Pressure sensors.
4. Pump operating data.
5. Coolant condition monitoring.
6. Model-based predictions.

The digital twin can estimate system state and identify deviations from expected behaviour.

18 Cybersecurity and Infrastructure Reliability

Modern cooling installations may depend on networked sensors, programmable logic controllers, supervisory control systems, and remote maintenance interfaces.

Their cybersecurity must therefore be included in system design.

Relevant principles include:

- Network segmentation.
- Strong authentication.
- Least-privilege access.
- Signed software updates.
- Secure logging.
- Backup and recovery.
- Offline safety controls.
- Independent emergency shutdown procedures.

A cooling system should retain safe operating behaviour when communications fail.

A simplified reliability representation for redundant components is:

$$R_{\text{parallel}} = 1 - (1 - R_1)(1 - R_2), \tag{125}$$

assuming independent component failures.

Real systems must account for common-cause failures, shared power supplies, common control systems, and correlated environmental conditions.

The security objective is protection of availability, integrity, and safety.

19 Philosophy, Logic, and Scientific Method

19.1 The distinction between a property and an outcome

A coolant has material properties, but its engineering performance is relational.

Thermal conductivity alone does not determine system heat removal. Viscosity alone does not determine pump energy. Specific heat alone does not determine total heat-exchanger effectiveness.

The relevant outcome emerges from the interaction between fluid, geometry, operating conditions, materials, and controls.

This distinction prevents the category error of treating an intrinsic property as a complete system-performance measure.

19.2 Scientific propositions

The theory can be organised around the following propositions.

Proposition A: Coolant performance is conditional on the system boundary and operating conditions.

Proposition B: No single thermophysical property is sufficient to determine overall cooling-system efficiency.

Proposition C: Different chemical formulations may be thermohydraulically equivalent under specified conditions.

Proposition D: Thermohydraulic equivalence does not imply chemical, electrical, environmental, or lifecycle equivalence.

Proposition E: The optimal coolant is application-dependent and may be a Pareto-efficient compromise rather than a universal maximum.

These propositions are testable through analytical derivation, simulation, and controlled experimentation.

19.3 Epistemic discipline

The research distinguishes:

1. Mathematical identity.
2. Physical law.
3. Empirical correlation.
4. Engineering approximation.
5. Statistical association.
6. Causal conclusion.
7. Commercial claim.

Each has a different evidential status.

A manufacturer's typical property value is not a universal physical constant. An empirical correlation is not a thermodynamic identity. A high coefficient of determination does not establish causal validity.

19.4 Ethics and social responsibility

The objective of cooling engineering is not simply to maximise machine performance.

A complete evaluation must consider:

- Worker health.
- Public safety.
- Environmental externalities.
- Water and energy consumption.
- Chemical handling and disposal.
- Reliability of essential infrastructure.
- Fair access to productive infrastructure.

Religious and philosophical traditions may provide ethical perspectives on stewardship, restraint, duty, and the responsible use of resources. Such perspectives can inform normative decisions, but they do not replace experimental thermodynamics or engineering validation.

20 Linguistics, Terminology, and International Standards

20.1 Terminological consistency

International research requires consistent technical terminology.

The database should preserve the manufacturer's original product designation while maintaining standardised technical fields.

Examples include:

- Specific heat capacity.
- Thermal conductivity.
- Dynamic viscosity.
- Kinematic viscosity.
- Heat-transfer coefficient.
- Overall heat-transfer coefficient.
- Vapour pressure.
- Freezing point.
- Bubble point.
- Dew point.

Translations should not alter chemical names, concentration bases, test methods, or engineering units.

20.2 International standards

The research should consult relevant standards from ASTM International, ISO, ASME, and other recognised standards bodies.

The Open Compute Project's PG25 specification is particularly relevant to data-centre cold-plate applications.

A standardised specification creates a more meaningful basis for comparison, but it does not guarantee that every conforming fluid has identical thermal performance.

21 Political Economy and Geopolitical Considerations

21.1 Cooling supply chains

Coolants depend on supply chains for:

- Base chemicals.
- Additives and inhibitors.
- High-purity water.
- Specialty fluids.
- Pumps and heat exchangers.
- Sensors and control systems.
- Industrial manufacturing capacity.

A disruption in one component may affect the availability of an entire cooling system.

21.2 Supply concentration

Let s_i be supplier i's share of a relevant coolant market.

The Herfindahl–Hirschman Index is:

$$HHI = \sum_{i=1}^{N} s_i^2. \tag{126}$$

A higher value indicates greater market concentration under the usual interpretation.

However, a market-share index should be complemented by analysis of geographic concentration, substitutability, technical qualification, and switching costs.

21.3 Strategic resilience

A resilient procurement strategy may include:

1. More than one technically qualified supplier.
2. Regional manufacturing capability.
3. Verified substitute formulations.
4. Adequate inventory for critical installations.
5. Documented compatibility and change-control procedures.
6. Secure access to replacement components.

Diplomacy and international cooperation can improve standards interoperability, supply-chain transparency, technical exchange, and crisis coordination.

The objective is to reduce unnecessary strategic dependence while preserving the benefits of international scientific and industrial collaboration.

22 Experimental Design and Validation

22.1 Laboratory requirements

A proper experimental programme should measure:

- Mass flow.
- Volumetric flow.
- Inlet and outlet temperatures.
- Pressure drop.
- Pump electrical power.
- Heat-exchanger duty.
- Coolant concentration.
- Coolant condition over time.

Temperature sensors must be calibrated and placed appropriately. Flow meters must be selected for the fluid's viscosity and expected flow regime.

22.2 Controlled comparison

Candidate coolants should be compared under identical:

1. Test geometry.
2. Heat input.
3. Flow constraint.
4. Inlet temperature.
5. Pressure boundary conditions.
6. Surface condition.
7. Measurement procedure.

Where identical conditions are impossible, the differences must be incorporated into the model.

22.3 Repeatability

For each test condition, conduct repeated measurements.

A simple mean is:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i. \tag{127}$$

The sample variance is:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2. \tag{128}$$

Confidence intervals should reflect the measurement design and the relevant statistical assumptions.

22.4 Long-duration testing

Short-duration thermal performance is insufficient for commercial selection.

Long-duration tests should examine:

- Oxidation.
- Thermal degradation.
- Corrosion.
- Inhibitor depletion.
- Contamination.
- Fouling.
- Seal compatibility.
- Material ageing.

The resulting dataset can support a lifecycle model rather than merely a short-run performance ranking.

23 Research Programme

23.1 Stage I: Manufacturer database

The initial target is at least 30 commercial formulations from at least 10 manufacturers.

Priority sources include:

- Dow.
- Eastman.
- Fleetguard.
- Clariant.
- BASF.
- Shell.
- Castrol.
- Arteco.
- Dynalene.
- PFX Group.

The target is a research objective, not a claim that 30 complete datasets have already been collected.

23.2 Stage II: Property estimation

Estimate temperature-dependent functions for:

$$c_p(T), \quad k(T), \quad \rho(T), \quad \mu(T). \tag{129}$$

Use alternative functional forms and compare their predictive accuracy.

23.3 Stage III: Phase and stability envelopes

Construct temperature-pressure-composition operating envelopes.

Overlay chemical stability, compatibility, and safety constraints.

23.4 Stage IV: Thermohydraulic performance

Calculate:

$$Re = \frac{\rho v D}{\mu}, \tag{130}$$

$$Pr = \frac{\mu c_p}{k}, \tag{131}$$

$$Nu = \frac{hD}{k}, \tag{132}$$

$$\Delta P = f_D \frac{L}{D} \frac{\rho v^2}{2}, \tag{133}$$

$$P_{\text{pump}} = \frac{\Delta P \dot{V}}{\eta_p}. \tag{134}$$

Use correlations appropriate to each flow regime and geometry.

23.5 Stage V: Equivalence analysis

Construct equivalence groups based on:

- Heat-transfer duty.
- Pressure drop.
- Pumping power.
- Operating temperature.
- Phase stability.
- Chemical compatibility.

Report the conditions and tolerances defining each group.

23.6 Stage VI: Economic optimisation

Calculate lifecycle costs, net present value, and levelised cost of cooling.

Assess sensitivity to electricity prices, fluid prices, discount rates, maintenance costs, and failure probabilities.

23.7 Stage VII: Experimental validation

Select representative candidates from each equivalence group for controlled laboratory testing.

The final objective is to test whether the predicted equivalence relationships survive experimental validation.

24 Limitations and Further Theoretical Development

The framework has several limitations that must be acknowledged.

1. Manufacturer datasets differ in temperature ranges, concentration bases, test methods, and publication quality.
2. Some products do not publish all four principal thermophysical properties.
3. Correlations may not transfer across geometries or flow regimes.
4. Mixture phase behaviour may require specialised thermodynamic models.
5. Chemical stability and corrosion require dedicated experiments.

6. Commercial formulations may change over time.
7. Product qualification is application-specific.
8. A fluid's thermal performance does not establish its environmental or occupational safety.

Further research should extend the model to:

- Non-Newtonian fluids.
- Nanofluids, with careful stability and safety assessment.
- Liquid-metal coolants.
- Refrigerants.
- Supercritical fluids.
- Two-phase cold-plate systems.
- Thermal energy storage materials.
- Hybrid cooling architectures.

These categories should not be forced into one empirical ranking. Their governing physics and operating constraints differ.

25 Conclusion

Heat is an unavoidable physical consequence of productive activity, but the economic cost of heat is not fixed.

Cooling engineering determines how efficiently thermal energy can be absorbed, transported, redistributed, recovered, and rejected. The performance of a coolant is determined not by one isolated material property but by the interaction of thermophysical properties, flow conditions, geometry, heat-exchanger design, chemical stability, materials compatibility, and operational controls.

This treatise has established a general mathematical framework beginning with the enthalpy balance and extending through heat-transfer correlations, hydraulic resistance, phase equilibrium, thermohydraulic equivalence, and economic optimisation.

The equivalence analysis is especially important. It demonstrates that different chemical formulations may produce comparable thermal and hydraulic outcomes under specified conditions. This creates the possibility of systematic coolant substitution, procurement diversification, and lifecycle optimisation.

However, thermohydraulic equivalence does not imply equivalence in corrosion protection, electrical properties, toxicity, environmental burden, or long-term reliability. These dimensions must remain explicit in the decision model.

The proposed Coolant Observatory will turn this theoretical framework into an empirical research platform. Manufacturer specifications will provide the initial data, statistical models will estimate temperature-dependent properties, engineering models will calculate system performance, and controlled experiments will test the resulting predictions.

The ultimate objective is a general, reproducible, and economically meaningful science of cooling: one capable of identifying not merely which coolant absorbs the most heat, but which coolant provides the most reliable, safe, efficient, and sustainable cooling service for a given productive system.

Acknowledgements and Research Scope

This treatise integrates established scientific and engineering principles with a proposed research architecture. The mathematical derivations are presented as theoretical results or conditional engineering models. The empirical tables reproduce selected manufacturer-published observations. The proposed database, equivalence classes, and optimisation platform require further implementation and validation.

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Glossary

Adiabatic Process
A thermodynamic process in which no heat is transferred across the system boundary.

Bubble Point
The temperature or pressure at which the first vapour bubble forms from a liquid mixture under specified conditions.

Coolant
A fluid or material used to absorb, transport, or redistribute thermal energy.

Dynamic Viscosity
A measure of a fluid's resistance to shear deformation.

Enthalpy
A thermodynamic state function defined as internal energy plus pressure-volume energy.

Equivalence Class
A set of objects satisfying a specified equivalence relation. In this treatise, it refers to coolants with comparable performance under explicitly defined operating conditions.

Heat Capacity
The amount of heat required to change the temperature of a specified quantity of material.

Heat-Transfer Coefficient
A coefficient describing convective heat transfer between a surface and a fluid.

Hydraulic Resistance
The relationship between flow and pressure loss in a fluid system.

Kinematic Viscosity
Dynamic viscosity divided by density.

Latent Heat
Thermal energy absorbed or released during a phase transition without the corresponding sensible temperature change.

Logarithmic Mean Temperature Difference
A mean temperature driving force used in heat-exchanger analysis.

Nusselt Number
A dimensionless measure of convective heat transfer relative to conduction.

Phase Envelope
The set of thermodynamic conditions delimiting phase behaviour in a substance or mixture.

Prandtl Number
A dimensionless ratio of momentum diffusivity to thermal diffusivity.

Reynolds Number
A dimensionless ratio characterising the relative importance of inertial and viscous forces in fluid flow.

Specific Heat Capacity
The heat required to raise the temperature of a unit mass of a substance by one kelvin.

Thermal Conductivity
A material property describing its ability to conduct heat.

Thermohydraulic Performance
The combined thermal and hydraulic behaviour of a cooling fluid and its associated system.

Thermophysical Property
A measurable physical property related to the thermodynamic or transport behaviour of a material.

Overall Heat-Transfer Coefficient
A coefficient representing the combined thermal resistance of the heat-transfer path.

Vapour Pressure
The equilibrium pressure exerted by a vapour in contact with its liquid or solid phase at a specified temperature.

Wetted Materials
The surfaces and components in direct contact with the coolant.

The End